

Aashish Awadhani

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SUMMARY

With experience in machine learning, I've successfully delivered real-time computer vision and LLM solutions. I led the development of OCR pipelines for CAPTCHA recognition, significantly improving sequence accuracy even in noisy inputs. Additionally, I engineered a real-time object detection stack on Jetson AGX Orin, which reduced latency by up to 40% and stabilized FPS for autonomous navigation. I'm eager to leverage my expertise in end-to-end ML pipelines to drive scalable AI products.

EXPERIENCE

CalSys Lab, Cal Poly Pomona | *AI/ML Research Assistant*

Sep 2025 - Present

- Built and benchmarked 4 OCR pipelines (CNN, YOLO-based detection, CRNN-CTC, Transformer OCR) for CAPTCHA recognition, improving sequence accuracy across noisy inputs.
- Developed character recognition models using PyTorch and TensorFlow, employing CNNs and Vision Transformers (ViTs) that increased recognition accuracy on distorted characters
- Improved handling of overlapping characters using a hybrid YOLO + CRNN architecture, reducing segmentation errors across complex datasets.

Autonomous Vehicle Lab, Cal Poly Pomona | *Machine Learning Engineer*

May 2024 - May 2025

- Engineered a real-time object detection stack using PyTorch + TensorRT on Jetson AGX Orin, achieving 35–40% lower latency and more stable FPS in field tests.
- Integrated YOLOv8 with ROS2 to enhance autonomous navigation using multi-sensor perception (RGB, LiDAR, depth).
- Performed model compression via quantization, pruning, and ONNX/TensorRT optimization, enabling reliable deployment under embedded hardware constraints.
- Used MLflow to track experiments and iteratively tune models based on realworld sensor data, which reduced model retraining time and improved detection accuracy

EDUCATION

California State Polytechnic University, Pomona | *Master of Science, Computer Science*(GPA: 3.97)Pomona Aug 2023 - May 2025

PROJECTS

InsightEngine AI

- Engineered InsightEngine AI, a document intelligence tool powered by Streamlit and Google Gemini 1.5/2.5 Flash, facilitates interactive, “chat-first” querying of intricate PDF documents such as business reports and research papers.
- Built a secure end-to-end document processing pipeline using PyPDF2 for text extraction and a privacy-centric session architecture that requires secure API keys for real-time analysis and summarization.

Real-Time Chat Application

- Built a responsive chat platform using MongoDB, Express.js, React, Node.js, and Socket.io with JWT-based authentication and global state via Zustand.
- Deployed real-time messaging infrastructure with WebSocket handling and clean UI using Tailwind CSS.

Stock Price Prediction with XGBoost

- Developed a time-series forecasting model with XGBoost, achieving 94% accuracy using advanced feature engineering (RSI, MACD, Bollinger Bands).
- Built a complete ML pipeline in Python with cross-validation and visualized trends using Matplotlib.

CERTIFICATION

- **Rapid Application Development with Large Language Models (LLMs):**NVIDIA Deep Learning Institute; Built LLM-based applications using Transformers, LangChain, and vector databases.; Developed text-generation, RAG, and semantic search pipelines for production-style use cases.

SKILLS

- **Machine Learning & AI:** PyTorch, TensorFlow, TensorRT, YOLO, OpenCV, CNN, RNN, Vision Transformers, Transformers, ONNX, Quantization, Pruning, CUDA, MLflow, Large Language Models, Hugging Face, LangChain, RAG, Vector DB, Natural Language Processing, Generative AI, Diffusion Models, AI Agents, OpenAI API, LLMOps, APIs
- **Software Engineering & Systems:** Python, C++, Java, JavaScript, Node.js, Express.js, Rest APIs, Docker, Git, Linux, ROS2, Fast API, MySQL, PostgreSQL, MongoDB, Redis, CI/CD, Streamlit, PyPDF2
- **Cloud & Tools:** AWS, GCP, Azure, GitHub Actions, PowerBI, Tableau